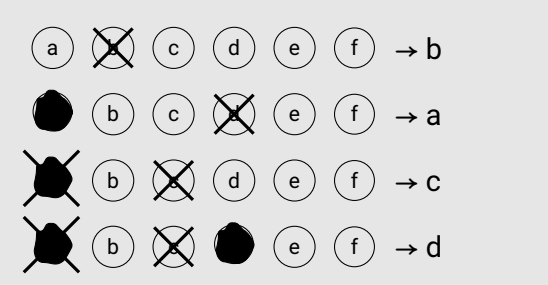
Exercises

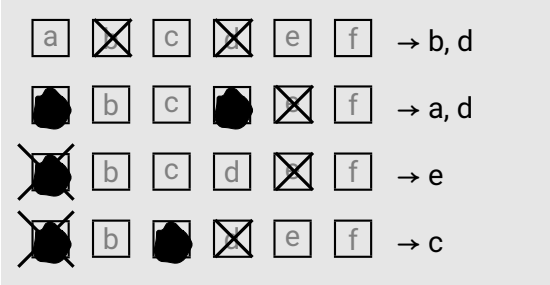
1	2	3	4	5	6

Surname, First name

1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	5	5	5	5	5	5	5
6	6	6	6	6	6	6	6
7	7	7	7	7	7	7	7
8	8	8	8	8	8	8	8
9	9	9	9	9	9	9	9
0	0	0	0	0	0	0	0

Machine Learning 1 (52041MAL6Y)
Midterm Exam





Answer multiple-choice questions as shown in the example. Circular checkboxes have only one correct answer. Square checkboxes may have multiple correct answers.

Midterm Exam for Machine Learning 1 - 25 September 2025

Pay attention to the following;

- **WRITE YOUR NAME AND STUDENT NUMBER ON THE FRONT PAGE** (don't forget to mark the digits)
- Write all your answers on this exam booklet; it will be scanned and digitally graded.
- Write your **answers inside the boxes** (it is ok, to slightly go over the margins though).
- The answer boxes should be large enough for your answer
- **If you need to empty the answer box in order to start over, ask the invigilator for a blank sticker or use the *Emergency Box* at the end of the booklet.**

During the exam you are allowed to use:

- The provided cheat sheet.
- Basic calculator (although you will not need it)

Please scan over the questions before you start to get an impression of the content

- You have 2 **hours** for the exam (except for those with pre-approved extensions)
- In total **28 points** can be earned divided over 5 exercises:
  - Exercise 1: Multiple choice questions [5 p.]
  - Exercise 2: True/False statements [5 p.]
  - Exercise 3: Classification [7 p.]
  - Exercise 4: Bayesian Linear Regression with Laplace prior [4 p.]
  - Exercise 5: Modeling wind speed with Maximum Likelihood Estimation [7 p.]
- We work with a partial grading: you can get points even if you don't manage to solve the full question.

Multiple choice questions

NOTE: For questions 1a and 1b, there is **only one** correct answer. Questions 1c and 1d have **multiple** correct options (you need to mark all correct options to get full score).

1p **1a** Consider two regression models estimated using regularized least squares

$$\hat{\mathbf{w}} = \frac{\beta}{2} \underset{\mathbf{w}}{argmin} \sum_{i=1}^{N} (y(x_i, \mathbf{w}) - t_i)^2 + \frac{\alpha}{2} \mathbf{w}^T \mathbf{w}$$

Model 1 has $\alpha = 0$, Model 2 has $\alpha > 0$. Which model is more likely to fit the training data well? Which model is more likely to fit the test data well?

(a) Training set: Model 1; Test set: Model 2.

(b) Training set: Model 1; Test set: impossible to tell.

(c) Training set: Model 2; Test set: Model 1.

(d) Training set: Equally likely, Test set: Model 1.

1p **1b** We want to minimize the mean squared error loss for a univariate linear regression model

$$E(w) = \frac{1}{N} \sum_{i=1}^{N} (w x_i - y_i)^2$$

using stochastic gradient descent with learning rate $\eta = 0.1$.

Suppose we have the dataset

$$(x_1, y_1) = (1, 2), \quad (x_2, y_2) = (2, 4),$$

and the initial parameter is $w^{(0)} = 0$.

In the first SGD step we sample the point $(x_2, y_2) = (2, 4)$. What is the value of the updated parameter $w^{(1)}$?

(a) $w^{(1)} = 1.0$

(b) $w^{(1)} = -1.0$

(c) $w^{(1)} = 1.6$

(d) $w^{(1)} = -1.6$

1.5p **1c** We consider two ways to solve a binary classification problem: linear regression and logistic regression. *Which of the following statements are correct?*

- ☐ Linear regression is more sensitive to outliers.
- ☐ Logistic regression uses a *linear model* for predicting the log-odds between the probabilities.
- ☐ Logistic regression is a probabilistic generative model because we can use the estimated probability distribution to sample new datapoints.
- ☐ We can use Gradient Descent in combination with the Binary Cross Entropy loss to estimate the Logistic regression model.

1.5p **1d** You are training a regression model on a dataset of size $N = 1000$. You decide to use k-fold cross-validation with $k = 10$. *Which of the following statements are correct?*

- ☐ Each sample is used exactly once for validation and $k - 1$ times for training.
- ☐ Each model in the cross-validation procedure is trained on the whole dataset (1000 samples) and tested on 100 samples.
- ☐ If this cross-validation setting is used to perform hyperparameter search over a grid with two parameters—one taking 3 possible values and the other taking 4 possible values, there will be 120 training runs.
- ☐ Increasing the number of folds k generally reduces the bias of the performance estimate but increases its variance.

True/False statements

NOTE: **Please write down a justification for your answer in each of the following questions**. Just writing True/False will not give you any points.

2p **2a** The function

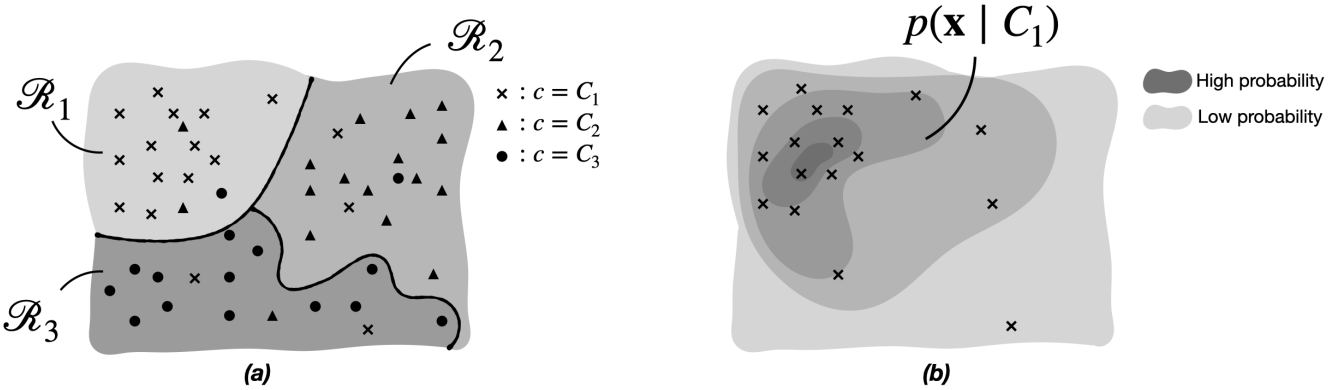
$$f(x) = \begin{cases} \frac{3}{4}(1-x^2), & -1 \le x \le 1, \\ 0, & \text{otherwise} \end{cases}$$

is a valid probability density function of a continuous random variable.

1.5p **2b** The Naïve Bayes assumption in a probabilistic classification model causes the number of parameters to increase compared to a fully general probabilistic model (without the Naïve Bayes assumption).

1.5p **2c** We consider a logistic regression model for classification and a test dataset. The model correctly classifies all samples in the test set (accuracy = 100%). The cross-entropy loss for the test dataset is thus zero.

Classification



Consider a three class classification problem with points $\mathbf{x} \in \mathbb{R}^2$ that belong to one of three classes $c \in \{C_1, C_2, C_3\}$. You have build a classifier by partitioning the 2D plane into three regions $\mathcal{R}_1$, $\mathcal{R}_2$, and $\mathcal{R}_3$ corresponding to the three classes (figure **(a)**). I.e., you say a datapoint $\mathbf{x} \in \mathbb{R}^2$ belongs to class C_2 if $\mathbf{x} \in \mathcal{R}_2$.

You assume that the actual observations follow class-conditional distributions $\mathbf{x} \sim p(\mathbf{x} \mid c = C_k)$, for which we may simply write $\mathbf{x} \sim p(\mathbf{x} \mid \mathcal{C}_k)$. So depending on which class the sample belongs to, it may appear more frequently in some regions than others (figure **(b)**).

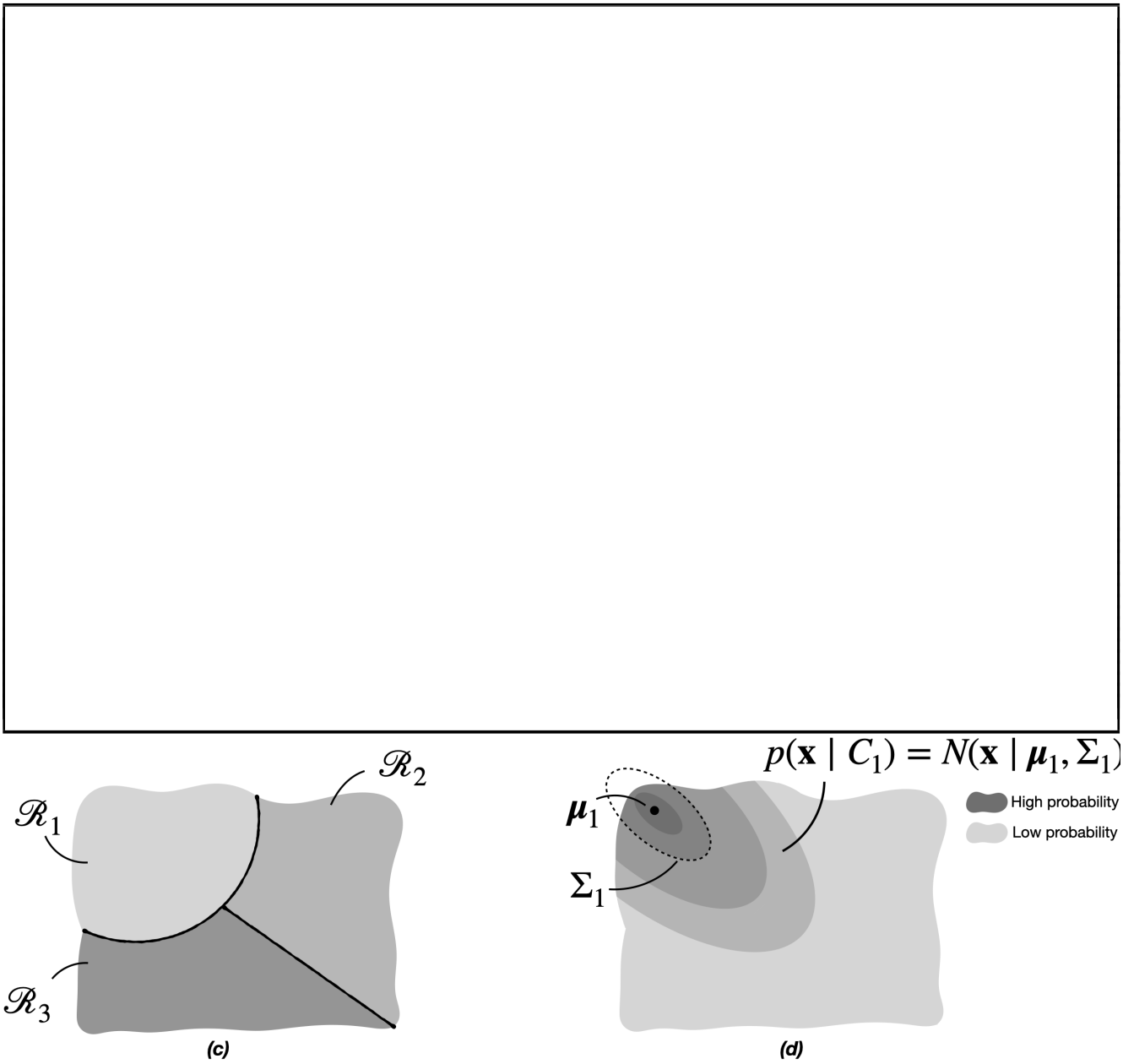
1p **3a** Suppose you know the class-conditional distribution $p(\mathbf{x} \mid c = \mathcal{C}_1)$. Give the probability that a point sampled from it is correctly classified as class $\mathcal{C}_1$. I.e., write out: $p(\mathbf{x} \in \mathcal{R}_1 \mid \mathcal{C}_1) = \ldots$

1p **3b** Without knowing anything about the class-conditional distributions of the other classes you can define the *total error rate* that your classifier makes on class $\mathcal{C}_1$. Let the total error rate be defined as the fraction of times a point $\mathbf{x} \sim p(\mathbf{x} \mid \mathcal{C}_1)$ is assigned any other class than its ground truth label $\mathcal{C}_1$. Please write down the expected error rate for points in class $\mathcal{C}_1$ based on $p(\mathbf{x} \mid \mathcal{C}_1)$.

1p **3c** All forms of errors, and the frequencies in which they are expected to occur, can be characterized in a so-called *expected* confusion matrix. Please give a definition of the expected confusion matrix in terms of the three class conditional distributions and the decision regions $\mathcal{R}_1$, $\mathcal{R}_2$, and $\mathcal{R}_3$.

1p **3d** For the purpose of classifying points into the three classes, instead of checking in which region $\mathcal{R}_k$ a point falls, one could also evaluate probabilities of a point belonging to that class $\mathcal{C}_k$, given that you also know the priors $p(C_k)$. Give the expression: $p(\mathcal{C}_k \mid \mathbf{x}) = \ldots$

1p **3e** Suppose you know all true class-conditional distributions, how would you define "optimal" decision regions? The best possible decision region $\mathcal{R}_i$ for class $\mathcal{C}_i$ is one such that...



We are going to assume that the data is randomly sampled according to the following probabilistic model. Each class-conditional is assumed to be given by a normal distribution (e.g. as in figure **(d)**) with mean $\boldsymbol{\mu}_k$ and covariance matrix Σ_k, i.e.,

$$p(\mathbf{x} \mid \mathcal{C}_k) = \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \Sigma_k)\,,$$

and the prior probability for each class is given by the values π_k, with $\sum_{k=1}^{3} \pi_k = 1$. I.e.,

$$p(C_k) = \pi_k\,.$$

1p **3f** Consider a dataset of points $\mathcal{D} = \{(\mathbf{x}_n, c_n)\}_{n=1}^{N}$ of N observed pairs of $\mathbf{x} \in \mathbb{R}^2$ and corresponding true class $c_n \in \{C_1, C_2, C_3\}$. Write down the likelihood of observing $\mathcal{D}$ under the above model with parameters $\{\boldsymbol{\mu}_k\}$, $\{\Sigma_k\}$, $\{\pi_k\}$. *Hint: make use of one-hot encoding* $z_{nk} = 1$ *if* $c_n = C_k$ *and* 0 *otherwise.*

1p **3g** Suppose the optimal decision boundaries are known as illustrated in figure **(c)**. Looking at the decision boundaries, what can we say about the covariance matrices?

Bayesian Linear Regression with Laplace prior

Consider the linear regression model

$$\mathbf{y} = \mathbf{X}\mathbf{w} + \boldsymbol{\varepsilon}, \qquad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n),$$

where:

- $\mathbf{X} \in \mathbb{R}^{N \times D}$ is a data matrix consisting of N i.i.d. samples.
- $\mathbf{y} \in \mathbb{R}^N$ is the vector of targets (including one scalar target for each of the N samples).
- $\mathbf{w} \in \mathbb{R}^D$ is the vector of coefficients. Each coefficient w_d, $d \in \{1, 2, ..., D\}$ has an independent Laplace prior

$$p(w_j) = \frac{\alpha}{2} \exp(-\alpha |w_j|), \qquad \alpha > 0.$$

3p **4a** Show that the optimization problem for the MAP estimator is

$$\hat{\mathbf{w}}_{\text{MAP}} = \arg\min_{\mathbf{w} \in \mathbb{R}^d} \frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \alpha \|\mathbf{w}\|_1.$$

↳

1p **4b** Which regularization technique is equivalent to the optimization problem that you just obtained?

Modeling wind speed via Maximum Likelihood Estimation

A meteorologist is analyzing wind speed measurements from a coastal weather station. The wind speed X (in m/s) follows a Rayleigh distribution, which has the following probability density function:

$$p(x \mid \theta) = \frac{x}{\theta^2} \exp\left(-\frac{x^2}{2\theta^2}\right), \quad x \geq 0, \theta > 0.$$

The meteorologist gathers a set of independent wind speed measurements $\mathbf{x} = \{x_1, x_2, ..., x_N\}$.

2p **5a** Compute the log-likelihood of the data $\mathbf{x}$ under the parameter θ.

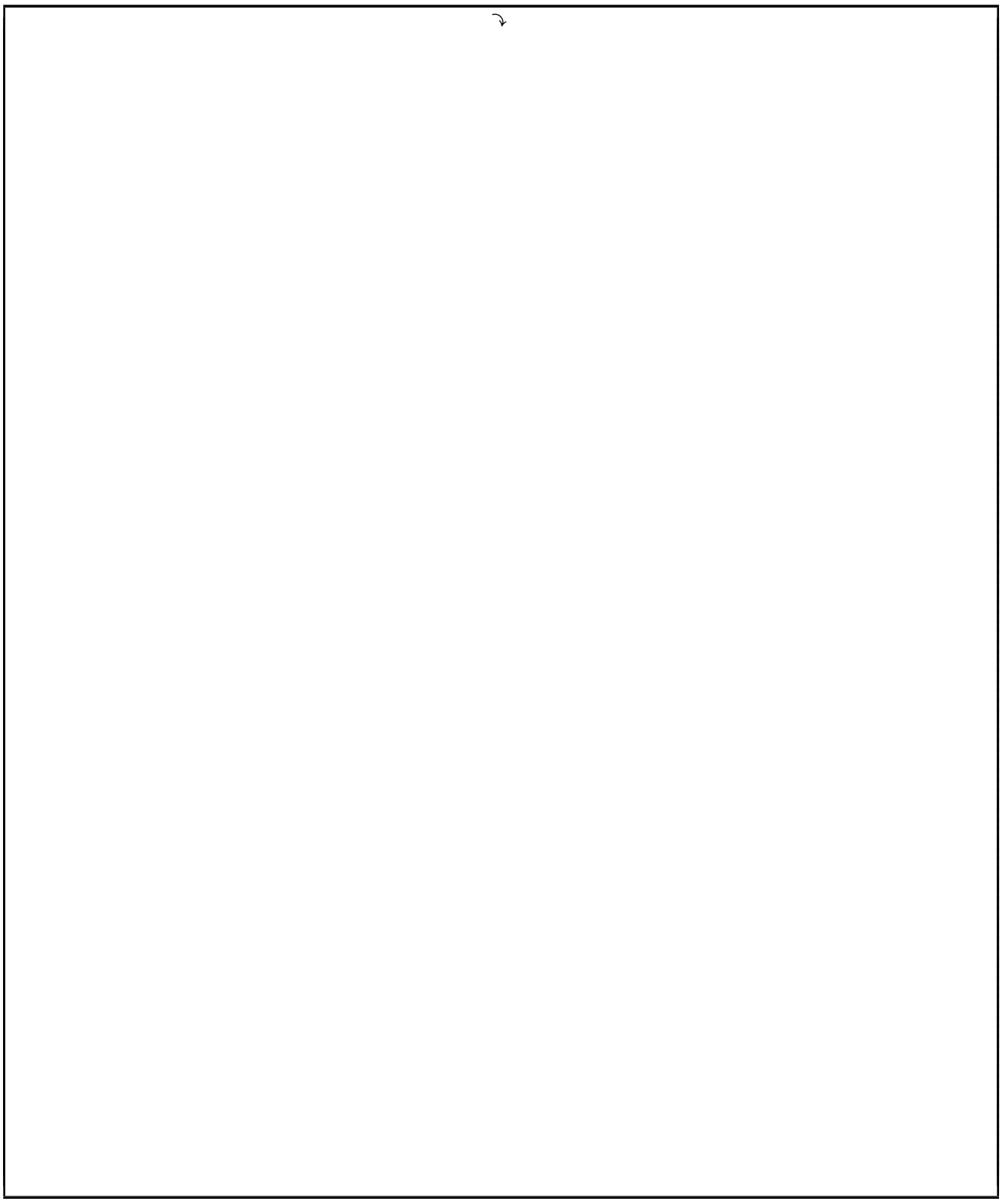
3p **5b** Compute the maximum likelihood estimator for θ^2.

2p **5c** Prove that the estimator that you have computed in question b) is unbiased.
Hint: You can use the following result:

If $X \sim$ *Rayleigh*(θ)*, then* $\mathbb{E}[X^2] = 2\theta^2$.

Emergency Box

1p **6** *Use this space only as a last resort* to provide answer to a previous question that didn't fit in the corresponding box. Indicate which question it is about (and also indicate it in the answer box of the corresponding question).



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